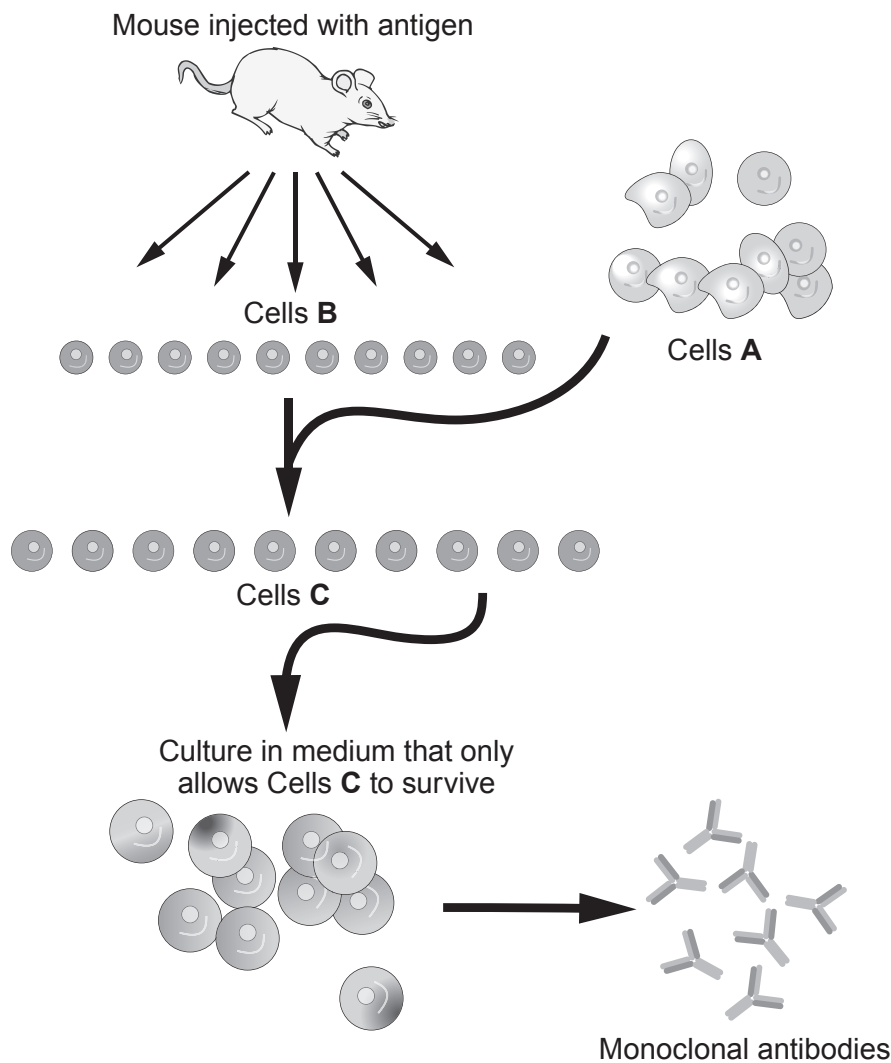


4. Basiliximab is a monoclonal antibody used to suppress the immune system of kidney transplant patients. It works by preventing white blood cells from attacking the cells of donor organs and therefore reduces the probability of rejection.

(a) (i) Apart from rejection, state another disadvantage of kidney transplants. [1]

(ii) Explain why and how white blood cells of the recipient would attack the cells of the donor organ therefore leading to rejection. [3]

(b) The diagram below shows a process for producing monoclonal antibodies.



Examiner
only

Use the diagram and your own knowledge to answer the following questions.

State the name of the cells labelled:

(i) **A** [1]

(ii) **B** [1]

(iii) **C** [1]

(iv) State how the following cells were produced: [2]

B

C

(c) State **two other** examples of medical uses for monoclonal antibodies. [2]

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